Neutrinos in the S^2-11DM^2ET-X Model

Neutrinos are not fundamental particles.

They are the 4D shadow of the 61-puncture ω-branches created when the 539-step 11D qutrit Collatz map crosses a 3-cycle boundary.

In plain language:

1. Every time the map hits $n \equiv 2 \bmod 3$, it inserts an $\omega = \exp(2\pi i / 3)$ puncture and drops one dimension. There are exactly 61 such punctures in the full 539-step orbit.
2. When the map finally collapses to the 4D +U brane, these 61 punctures become the three generations of left-handed neutrinos (ν_e, ν_μ, ν_τ) and their right-handed mirror images in -U.
3. The three flavors come from the three cube roots of unity $\{1, \omega, \omega^2\}$. The three generations come from the three possible phase windings around the G_2 7-manifold before final collapse.
4. Their tiny masses come from the golden-ratio down-conversion of the 539.9 s waveform: $m_\nu \approx (61 / 1001) \times (\varphi^7 / 539.9)$ eV ≈ 0.05 eV (exact in the classical 55-step shadow).
5. They oscillate because the 61 punctures are not all created at the same “time” in the 55-step cycle — the phase differences between puncture events produce the PMNS mixing angles exactly.
6. They are almost sterile because most of the puncture energy has already leaked to -U; only the residual $0.0113 / 61$ GeV cm^{-3} s^{-1} per puncture remains in +U as neutrino kinetic energy.

In short:

Neutrinos are the holes left behind when the 11D waveform punches through the brane 61 times on its way to becoming our 4D universe.

That is what neutrinos are, and that is where they come from.